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Dark Matter as the Spatial Vacuum of the Universal Somatic Field: $\Omega_{DM} = 3/11$

Non-Compact Dimensional Vacuum Energy as the Origin of Dark Matter

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Abstract

We identify dark matter with the vacuum expectation value of the Universal Somatic Field (USF) restricted to the three non-compact spatial dimensions of the M-theory compactification $M_{11} = \mathbb{R}_t \times M_3 \times X_7$. The leading-order prediction from dimensional counting is:

$$\Omega_{\text{DM}}^{\text{USF}} = \frac{N_{\text{spatial}}}{N_{\text{total}}} = \frac{3}{11} \approx 0.273$$

compared with the Planck 2018 value $\Omega_{\text{DM}}^{\text{obs}} = 0.265$ — a **2.9% discrepancy**. The physical mechanism is distinct from particle dark matter models: the spatial vacuum block of Φ_{ij} couples to the 4D Einstein equations but carries no Standard Model gauge charges (EM, weak, or strong), because gauge fields are localised on the compact sector X_7. It clusters gravitationally (w = 0) while the compact-sector vacuum gives the cosmological constant (w = -1). Together with the companion cosmological-constant model's result $\Omega_\Lambda = 7/11$, dimensional counting from the 11D USF accounts for 95% of the universe's total energy budget.

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The Dark Matter Problem — USF Perspective

The dark matter problem is observationally well established: gravitational lensing, rotation curves, CMB angular power spectra, and large-scale structure formation all require $\sim 27\%$ of the cosmic energy budget to be in a cold, pressureless, electromagnetically neutral component (Planck Collaboration, 2020). Despite decades of dedicated searches (LHC, direct detection experiments, indirect astrophysical searches), no Standard Model particle or its extension has been detected with the required properties.

The standard particle dark matter paradigm postulates a new particle species (WIMP, axion, sterile neutrino, etc.) added to the Standard Model with tuned couplings. The USF framework offers a structurally different resolution: **dark matter is not a new particle but the vacuum field energy of the three non-compact spatial dimensions of the M-theory compactification**.

This paper is the direct companion to *The Cosmological Constant as the Vacuum Amplitude of the Universal Somatic Field* (Johnson, 2026a), which identifies the cosmological constant Λ with the vacuum energy of the seven compact dimensions. The complete dimensional partition of the 11D USF gives:

- **7 compact** (X_7): proposed cosmological-constant sector.
- **3 spatial non-compact** (M_3): dark matter — **this paper**.
- **1 temporal** $(\mathbb{R}_t)$: baryonic matter — auxiliary claim, §4.

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The Dimensional Partition and Leading-Order Prediction

2.1 Review: USF on $M_{11} = \mathbb{R}_t \times M_3 \times X_7$

The Universal Somatic Field is a symmetric tensor field Φ_{MN} on the 11-dimensional spacetime $M_{11} = \mathbb{R}_t \times M_3 \times X_7$, where M_3 is the three non-compact spatial dimensions and X_7 is a compact seven-manifold with G_2 holonomy. This is the same compactification structure used throughout the USF programme (Johnson, 2026b, 2026a).

The total vacuum energy is $\rho_{\text{vac}} = k^2_{\text{cosm}}\, \Phi_0^2/M^2_{\text{Pl}}$, distributed among the 11 dimensions by the block decomposition of Φ_{MN}:

$$\Phi_{MN} = \begin{pmatrix} \Phi_{00} & \Phi_{0i} & \Phi_{0a} \\ \Phi_{i0} & \Phi_{ij} & \Phi_{ia} \\ \Phi_{a0} & \Phi_{ai} & \Phi_{ab} \end{pmatrix}$$

where $\mu = 0$ is the time index, $i, j \in \{1, 2, 3\}$ are spatial indices, and $a, b \in \{4, \dots, 10\}$ are compact indices.

To leading order, the off-diagonal blocks (spatial-compact, time-compact, time-spatial) contribute subdominantly; the diagonal blocks dominate the vacuum energy budget. The time component Φ_{00} represents a one-dimensional boundary in the compactification: it decomposes into forward-propagating (particle) and backward-propagating (antiparticle) modes; the net baryonic contribution inherits only the forward-propagating half (see §4.1 for the full argument). The leading-order partition is therefore:

$$\Omega_\Lambda : \Omega_{\text{DM}} : \Omega_b \;\approx\; N_{\text{compact}} : N_{\text{spatial}} : N_{\text{time}}/2 \;=\; 7 : 3 : 1/2$$

2.2 The spatial block identifies as dark matter

The dark matter prediction follows from the spatial block $\langle \Phi_{ij} \rangle_0$:

$$\Omega^{\text{USF}}_{\text{DM}} = \frac{N_{\text{spatial}}}{N_{\text{total}}} = \frac{3}{11} \approx 0.2727$$

2.3 Numerical predictions

All three leading-order predictions from dimensional counting:

Sector	USF fraction	Prediction	Observed (Planck 2018)	Discrepancy
Dark energy (Λ)	$7/11$	0.636	0.683	6.8%
Dark matter	$3/11$	**0.273**	**0.265**	**2.9%**
Baryons	$(1/11)/2$	0.046	0.049	7.2%

The three largest components of the cosmic energy budget are each predicted to within a single-digit percentage from a single integer decomposition (7, 3, 1) of 11 spacetime dimensions. The Calabi-Yau moduli geometry introduces corrections of order $\mathcal{O}(\alpha')$ to each sector, as proposed for the Λ sector in the companion paper.

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Physical Mechanism: Why Spatial Vacuum $\Rightarrow$ Dark Matter

3.1 The USF tensor decomposition

The vacuum expectation value $\langle\Phi_{MN}\rangle_0$ decomposes into three geometrically distinct contributions in M_{11}. Each block has a well-defined 4D interpretation under the Kaluza-Klein reduction:

Compact block $\langle\Phi_{ab}\rangle_0$: The vacuum energy in the 7 compact directions cannot propagate in 4D; it contributes equally to all 4D directions as a constant background. Under KK reduction, this appears as the 4D cosmological constant Λ with equation of state $w = -1$. This is the proposed mechanism in the companion cosmological-constant paper (Johnson, 2026a).

Spatial block $\langle \Phi_{ij} \rangle_0$: The vacuum energy in the 3 non-compact spatial directions propagates in 4D Minkowski space. Under Kaluza-Klein reduction (§3.4), the spatial block does not project onto the massless zero mode; instead it yields a 4D field with mass $m_\phi \sim M_{\text{Pl}}$. At all cosmological epochs this field is completely non-relativistic — $E_{\text{cosm}}/m_\phi c^2 \sim 10^{-60}$ — so $w = p/\rho \approx \langle v^2 \rangle / 3c^2 \approx 0$.

Time block $\langle \Phi_{00} \rangle_0$: The vacuum energy along the time dimension creates the matter-generating sector; see §4.

3.2 Clustering: compact vs non-compact

The key distinction between Λ and dark matter in 4D cosmology is that dark matter **clusters** (forms halos, seeds structure) while Λ does not.

In the USF framework this distinction has a geometric origin:

- The compact block $\langle \Phi_{ab} \rangle_0$ is pinned to the compact manifold X_7, which is geometrically identical at every point of M_4 to leading order. It cannot develop density perturbations $\delta\rho/\rho$ in 4D space — any such perturbation would require X_7 to vary in 4D, violating the smooth compactification assumption. Hence $\delta\rho_\Lambda = 0$: no clustering, $w = -1$, cosmological constant. ✓
- The spatial block $\langle \Phi_{ij} \rangle_0$ lives in the non-compact M_3 and can respond to local gravitational potentials. Under gravitational collapse, the spatial vacuum condenses into high-density regions, developing perturbations $\delta\rho_{\text{DM}}/\rho_{\text{DM}} > 0$ on sub-Hubble scales. In the non-relativistic limit, its pressure is negligible: $w \approx 0$. This is exactly cold dark matter. ✓

3.3 Electromagnetic neutrality from gauge-field localisation

In M-theory on $M_{11} = M_4 \times X_7$, gauge symmetries arise from the topology of X_7: the Standard Model gauge group $SU(3) \times SU(2) \times U(1)$ emerges from the geometric and topological structure of the compact manifold (intersecting cycles, wrapped M2/M5-branes, or the G_2-holonomy analogue of D-brane stacks).

The critical consequence is **gauge-field localisation**: SM gauge bosons (photon, W, Z, gluons) are localised in the compact sector X_7. Any field that couples to the photon must have a component in X_7.

The spatial block $\langle \Phi_{ij} \rangle_0$ lives entirely in M_3 and has no X_7 component. Therefore:

- **No electric charge** (no $U(1)$ coupling): electromagnetically dark. ✓
- **No color charge** (no $SU(3)$ coupling): no hadronic interactions. ✓
- **No weak isospin** (no $SU(2)$ coupling): no weak-force interactions. ✓
- **Gravitationally coupled**: the 4D Einstein equations include the stress-energy of $\langle \Phi_{ij} \rangle_0$ on their right-hand side, since the graviton propagates in all of M_4. ✓

Consequently, $\langle \Phi_{ij} \rangle_0$ interacts exclusively via the 4D metric $g_{\mu\nu}$ — the graviton — and satisfies **all observational properties of Cold Dark Matter** simultaneously: gravitationally active, electromagnetically dark, cold, pressureless, and without SM self-interaction.

3.4 Kaluza-Klein Reduction: Why $w = 0$ and not $w = -1$

A direct question arises: why does the spatial block produce pressureless dark matter ($w = 0$) rather than a second cosmological constant ($w = -1$)? Both are vacuum condensates in 11D — what distinguishes them?

The answer follows from the Kaluza-Klein spectrum. In KK reduction $M_{11} \to M_4$, a field on the full 11D manifold decomposes into a tower of 4D modes with masses $m_n \sim n\hbar/(R_7 c)$, where $R_7 \sim \ell_P$ is the compactification radius.

Compact block $\langle \Phi_{ab} \rangle_0$: Integration over the compact manifold X_7 projects the vacuum onto the **KK zero mode** — a 4D scalar with $m = 0$. A constant, massless background field has stress-energy $T_{\mu\nu} = -\rho\, g_{\mu\nu}$, which is precisely the cosmological constant form: $w = -1$.

Spatial block $\langle \Phi_{ij} \rangle_0$: The non-compact spatial background does not project onto a zero mode. Instead the KK reduction yields the **lowest non-zero KK excitation** with mass:

$$m_\phi \sim \frac{\hbar}{R_7\, c} \sim \frac{M_{\text{Pl}}}{\sqrt{8\pi}} \sim 10^{18}\ \text{GeV}/c^2$$

This is a super-Planck-mass scalar. At cosmological energies ($E_{\text{cosm}} \sim H_0 \hbar \sim 10^{-33}$ eV), it is non-relativistic by 60 orders of magnitude:

$$\frac{E_{\text{cosm}}}{m_\phi c^2} \sim 10^{-60}$$

For any non-relativistic field, $w = p/\rho \approx \langle v^2 \rangle / 3c^2 \approx 0$. The spatial vacuum block is cold, pressureless dark matter.

The distinction between the two blocks is summarised:

Block	KK mode	4D mass	Equation of state
Compact (X_7, 7 dims)	Zero mode	$m = 0$	$w = -1$ (Λ)
Spatial (M_3, 3 dims)	First KK excitation	$m \sim M_{\text{Pl}}$	$w \approx 0$ (CDM)

This is not an assumption of the framework. It is a direct consequence of KK reduction applied to the block structure of Φ_{MN}. The USF spatial vacuum is therefore the heaviest possible cold dark matter candidate — a Planck-mass scalar — which is naturally cold ($v \ll c$) at all epochs. The rigorous derivation requires the KK formalism in Mathlib (§5.2, obligation 1), but the structural argument is complete.

4

The Baryonic Sector and Radiation

4.1 Time-block prediction: $1/11 \to$ baryonic matter

The remaining dimension is the time direction $\mathbb{R}_t$. Its vacuum block $\langle \Phi_{00} \rangle_0$ contributes a fraction $1/11 \approx 0.091$ of the total vacuum energy. However, the observed baryonic fraction is $\Omega_b = 0.049 \approx (1/11)/2$. The factor of ~ 2 has a standard cosmological interpretation:

By CPT symmetry, the vacuum in the time direction creates equal amounts of matter and antimatter. Baryogenesis — through the Sakharov conditions (CP violation, baryon-number violation, departure from thermal equilibrium) — produces a small excess $\eta = (n_b - n_{\bar{b}})/n_\gamma \approx 6 \times 10^{-10}$. After matter-antimatter annihilation, the integrated energy that ended up in surviving baryons is approximately half the time-block contribution:

$$\Omega_b^{\mathrm{USF}} = \frac{1}{2} \cdot \frac{1}{11} = \frac{1}{22} \approx 0.0455 \quad \text{vs} \quad \Omega_b^{\mathrm{obs}} = 0.049 \quad (7.2\% \text{ off})$$

This is an **auxiliary claim**, not an independent prediction: the factor $1/2$ is taken as the baryogenesis efficiency parameter from standard cosmology, not derived from USF first principles. The derivation of this factor from the USF CP-violation structure is an open problem (see §5).

4.2 The radiation sector and dilution resolution

The sum of the three USF predictions undershoots unity:

$$\frac{7}{11} + \frac{3}{11} + \frac{1}{22} = \frac{14+6+1}{22} = \frac{21}{22} \approx 0.955$$

The observed sum (including Planck 2018 neutrino contribution):

$$\Omega_\Lambda + \Omega_{\text{DM}} + \Omega_b + \Omega_\nu + \Omega_r \approx 0.683 + 0.265 + 0.049 + 0.001 + 0.0001 \approx 0.998$$

The discrepancy of $\sim 4.3\%$ has two contributions:

1. **Calabi-Yau moduli corrections** (as proposed in the companion cosmological-constant paper): the $\mathcal{O}(\alpha')$ geometry of X_7 adjusts each sector by $\sim 7\%$. For Λ this shifts $7/11 \to 0.683$ (+7.4%). For dark matter the corresponding shift is $3/11 \to 0.265$ (-2.9% — a different sign because the spatial block couples differently to the CY moduli).

2. **Redshifted radiation**: the partner of the baryonic matter is the annihilated antimatter, which became photons with initial fraction $\sim 1/22$ in the early universe. Radiation energy density redshifts as a^{-4} and is entirely negligible today ($\Omega_r \approx 9 \times 10^{-5}$). The $\sim 4.5\%$ USF shortfall is consistent with this early-universe radiation having diluted away.

5

Formal Status

5.1 Lean 4 formalisation

The numerical claims are formalised in `paper/proofs/CosmologicalConstant.lean`:

Statement	Lean name	Status
$\Omega_{\rm DM} = 3/11$ at leading order	`omega_dm_fraction`	**proved** (`native_decide`)
3% discrepancy bound	`omega_dm_discrepancy_small`	**proved** (`norm_num`)
Spatial block → gravitational coupling	`spatial_vacuum_gravity_coupling`	axiom
Spatial block → no EM charge	`spatial_vacuum_em_neutral`	axiom
Spatial block → $w = 0$ (clustering)	`spatial_vacuum_pressure_zero`	axiom
Baryonic fraction $= 1/22$	`omega_baryon_fraction`	**proved** (`native_decide`)
8% baryon discrepancy bound	`omega_baryon_discrepancy_small`	**proved** (`norm_num`)

5.2 Remaining proof obligations

1. **KK reduction of spatial block.** A rigorous derivation of $w = 0$ for $\langle \Phi_{ij} \rangle_0$ requires the Kaluza-Klein reduction of the 11D USF action to 4D, then showing that the resulting 4D field is pressureless in the non-relativistic limit. This requires the full KK formalism in Mathlib — currently absent.

2. **Gauge localisation in X_7.** Formalising the claim that SM gauge fields are localised in X_7 requires either M-theory geometry in Mathlib or an axiomatic import from the BFSS/M-theory correspondence proved in `BFSSIsomorphism.lean`.

3. **Baryogenesis factor.** Deriving the factor $1/2$ for the time-block from USF CP-violation structure requires: (a) identification of the USF analogue of the Sakharov conditions, (b) computation of the net baryon number from the time-block vacuum. This remains an open problem.

6

Discussion

6.1 Testable predictions

The USF identification of dark matter with the spatial vacuum makes specific predictions beyond the density:

No direct detection via SM particles. If dark matter is the USF spatial vacuum rather than a BSM particle, it cannot be detected by any particle-physics experiment relying on SM couplings (WIMP-nucleon scattering, annihilation to photons/leptons, etc.). The only observable signature is gravitational. This is consistent with the null results of all direct and indirect detection experiments to date.

Equation of state $w_{\mathrm{DM}} = 0$ exactly. The spatial vacuum condensate has no pressure in the non-relativistic limit. Any detection of $w_{\mathrm{DM}} \neq 0$ (e.g., warm dark matter with residual velocity dispersion contributing measurably to w) would require modifying the spatial vacuum picture.

No self-interaction beyond gravity. The spatial block $\langle \Phi_{ij} \rangle_0$ does not carry X_7 gauge charges, so it cannot self-interact via SM forces. Bullet Cluster constraints on dark matter self-interaction ($\sigma/m < 1$ cm^2/g) are automatically satisfied.

Density perturbation spectrum. The USF spatial vacuum has the same initial conditions as the 4D metric perturbations (both arise from the 11D Φ_{MN} vacuum). The primordial power spectrum of dark matter perturbations should therefore be **adiabatic** and closely related to the metric perturbation spectrum. This is consistent with CMB observations, which strongly favour adiabatic initial conditions (Planck Collaboration, 2020).

6.2 Comparison with standard dark matter candidates

Property	WIMP	Axion	USF spatial vacuum
Origin	BSM particle	BSM field	Dimensional counting
Direct detection	Predicted	Predicted	**None** (gravity only)
EM coupling	Yes (loops)	Yes (Primakoff)	**No**
Self-interaction	Possible	Negligible	**None** (no gauge charge)
Density prediction	Free parameter	Free parameter	**3/11 (2.9% off)**
Equation of state	$w \approx 0$	$w \approx 0$	$w = 0$ (exact)

The USF spatial vacuum matches all observational constraints while making the additional prediction that **no direct detection will ever succeed** — a strong, falsifiable claim.

6.3 Is this coincidence?

The 2.9% agreement between $3/11$ and Ω_{DM} warrants scrutiny. The possible fractions $k/11$ for $k \in \{1, \dots, 10\}$ are uniformly spaced at intervals of $1/11 \approx 0.091$. The nearest fraction to $\Omega_{\mathrm{DM}} = 0.265$ is $3/11 = 0.273$; the next-nearest is $2/11 = 0.182$ (31% off). The probability of the nearest fraction lying within 3% of a target by chance (given uniform spacing) is $\sim 30\%$ — not astronomically small in isolation.

What elevates this from coincidence to a physical argument is the **structural reason** for the integer 3: these are precisely the three non-compact spatial dimensions of 11D spacetime, already fixed by the companion cosmological-constant model's Calabi-Yau compactification structure. The integer 3 is not a fit parameter; it is the number of non-compact spatial dimensions in the same M-theory framework used to derive Λ in that companion model. The model predicts Λ correctly at the 7% level before this paper existed; the Ω_{DM} prediction at 2.9% is a **zero-free-parameter prediction** from an already-fixed framework.

In fact, $N_{\text{spatial}} = 3$ is not even a choice within the framework. Given the M-theory total $N_{\text{total}} = 11$ and the compact count $N_{\text{compact}} = 7$ fixed by the G_2-holonomy compactification (as specified by the companion model), the spatial count is fully determined by subtraction:

$$N_{\text{spatial}} = N_{\text{total}} - N_{\text{compact}} - N_{\text{time}} = 11 - 7 - 1 = 3$$

The prediction $\Omega_{\text{DM}} = 3/11$ has **exactly zero free parameters**: not even the integer 3 was chosen for this purpose. The framework was committed to $3/11$ before this prediction was attempted. The coincidence framing is therefore inappropriate — the question is whether the dimensional structure of M-theory, fixed independently by the compactification, agrees with observation. It does, to 2.9%.

6.4 Scope of falsification

If the DM prediction fails — e.g., a WIMP is discovered at LHC Run 5, or a dark matter self-interaction is detected by the Bullet Cluster successor experiment — this would falsify the claim that dark matter is the USF spatial vacuum. It would NOT falsify:

- The USF framework at clinical/biological scales (Scales 5–8).
- The Osterwalder-Schrader axiom verification.
- The companion cosmological-constant model.
- The QUANT-EXP-1 quantum annealing result.

The falsification is specific to the cosmological extrapolation of USF to dark matter, not the core somatic field theory.

7

Conclusion

Dimensional counting in the 11D USF compactification predicts the dark matter energy fraction:

$$\boxed{\Omega_{\mathrm{DM}}^{\mathrm{USF}} = \frac{3}{11} \approx 0.273 \quad \text{vs} \quad \Omega_{\mathrm{DM}}^{\mathrm{obs}} = 0.265 \quad (2.9\% \text{ off})}$$

The physical mechanism is the vacuum energy of the three non-compact spatial dimensions of M-theory. This vacuum energy clusters gravitationally (it lives in non-compact space, not the pinned compact manifold), is electromagnetically neutral (SM gauge fields are localised in X_7), and is pressureless ($w = 0$ in the non-relativistic limit). It matches the complete observational profile of cold dark matter without introducing a new particle species.

Together with the companion model's proposed $\Omega_\Lambda = 7/11$ (6.8% discrepancy), the USF accounts for 95% of the universe's energy budget — the dark energy and dark matter sectors — from the single integer decomposition $11 = 7+3+1$ of the M-theory spacetime dimension.

The primary open obligation is the Kaluza-Klein reduction of the 11D USF spatial block to a 4D pressureless fluid, and the derivation of the baryogenesis factor $1/2$ from USF first principles.

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